

A simple model predicts how warming simplifies wild food webs

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Warming increases the metabolic demand of consumers¹, strengthening their feeding interactions². This could alter energy fluxes^{3–5} and even amplify extinction rates within the food web^{6–8}. Such effects could simplify the structure and dynamics of ecological networks^{9,10}, although an empirical test in natural systems has been lacking. Here, we tested this hypothesis by characterizing around 50,000 directly observed feeding interactions across 14 naturally heated stream ecosystems^{11–15}. We found that higher temperature simplified food-web structure and shortened the pathways of energy flux between consumers and resources. A simple allometric diet breadth model^{10,16} predicted 68–82% of feeding interactions and the effects of warming on key food-web properties. We used model simulations to identify the underlying mechanism as a change in the relative diversity and abundance of consumers and their resources. This model shows how warming can reduce the stability of aquatic ecosystems by eroding the structural integrity of the food web. Given these fundamental drivers, such responses are expected to be manifested more broadly and could be predicted using our modelling framework and knowledge of how warming alters some routinely measured characteristics of organisms.

All natural systems contain complex food webs and their stability is shaped by non-random structural properties¹⁷. For example, the strength of consumer–resource interactions can influence stability^{8,19} and energy often flows from many abundant small species into progressively fewer large species, especially in the aquatic realm²⁰. Global warming could disrupt these patterns, yet we lack high-quality field data to test and validate predictive models of temperature effects on food webs. In theory, consumers should exert stronger feeding pressure on the biomass stocks of lower trophic levels in warmer environments^{3,4}, but may struggle to meet their rising energy demands^{7,8}. This could lead to shorter food chains⁶, simpler food webs⁹, less efficient energy flux⁵ and an altered distribution of biomass throughout the food web²¹.

To test these expectations, we used dietary analysis to comprehensively characterize food-web interactions for 14 geothermally heated streams in Iceland (see Methods). The streams occur within 1.5 km of each other in a pristine mountain landscape (Supplementary Fig. 1) that is free from anthropogenic influences other than the occasional grazing of sheep. The streams are very similar in their physical and chemical properties, yet vary in

temperature from 5 to 25 °C due to indirect heating of groundwater through the bedrock (Supplementary Tables 1–3). As the streams occur in the same catchment, they avoid the biogeographical differences that are associated with other natural gradients in temperature (for example, latitude or altitude)²². This study system thus acts as a space-for-time proxy in which temperature effects on food-web structure can be investigated in a wild setting with all of the complexity and realism of natural ecosystems²².

We used an allometric diet breadth model (ADBM)^{10,16}—parameterized with data on the average body mass and the population abundance of species sampled in each stream during August 2008 (that is, no a priori information on feeding links)—to predict the structure of each food web (see Methods). We then examined how several properties of the ADBM-predicted food webs varied with stream temperature and found significant linear relationships for four key metrics that are related to food-chain length, complexity, energy flux and biomass distribution (Supplementary Fig. 2). This allowed us to formulate four hypotheses (H1–H4) that could be tested with an empirical quantification of feeding links in the system. We anticipated that, as stream temperature increases, there should be: (H1) a reduction in mean trophic level; (H2) a decrease in connectance; (H3) shorter pathways of energy flux through the food web; and (H4) an increase in the biomass of consumers relative to their resources.

We tested our predictions by characterizing the actual food-web structure of each stream on the basis of almost 50,000 observations from gut contents (see Methods). There was a simplification of food-web structure as stream temperature increased, from a diffuse, reticulate network (Fig. 1a) to one with fewer and shorter chains (Fig. 1b). In support of H1, mean trophic level was lower in the warmer streams (Fig. 1c), with herbivorous interactions becoming increasingly dominant. This seemed to be driven by a disproportionate loss of consumer species relative to resources as the temperature of the stream increased (Supplementary Fig. 3). Consumer losses probably occurred as a result of consumers being unable to meet the greater metabolic demands of the warmer environment^{1,7} and/or withstand increased predation by brown trout, an apex predator that cannot persist in the coldest streams due to its own metabolic constraints^{13,14}. Warmer food webs were also less connected (Fig. 1d)—as predicted in H2—suggesting that they will be more sensitive to secondary extinctions^{23,24} and dominated by more specialized consumers, with energy channelled through fewer and

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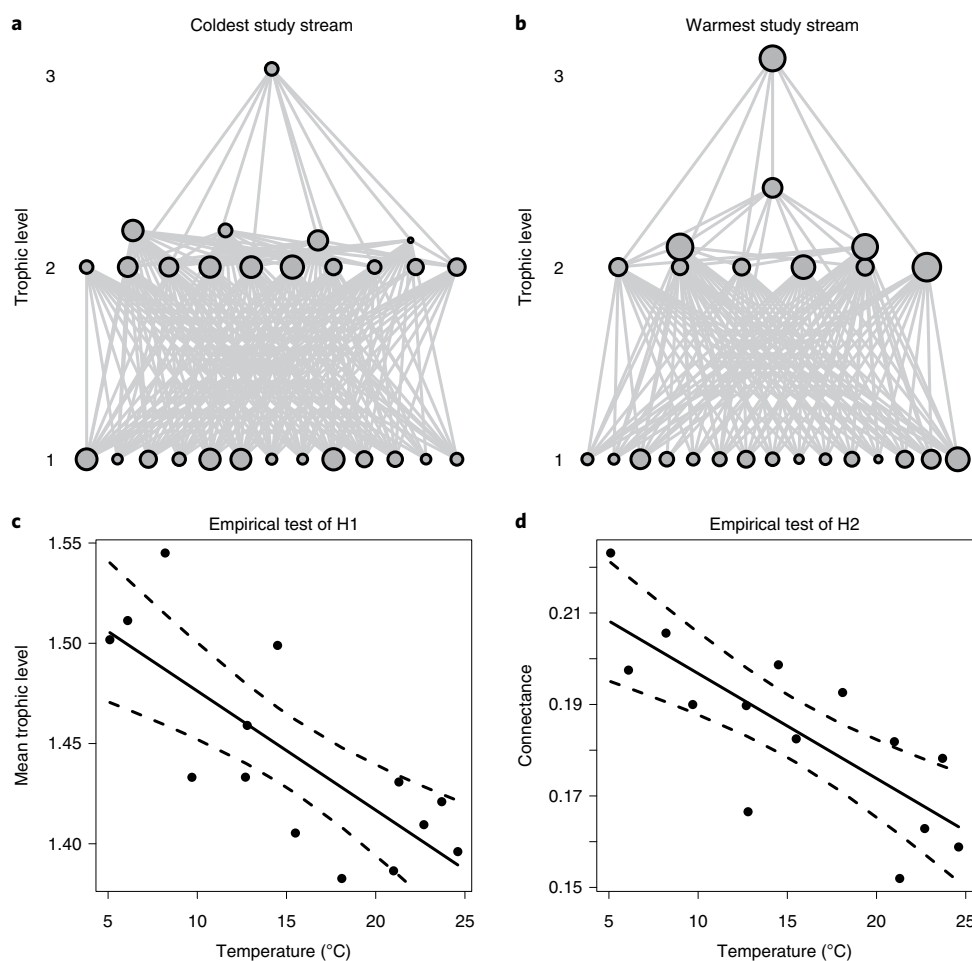


Fig. 1 | The effect of temperature on the properties of food webs. a,b, Food webs for the coldest (**a**) and warmest (**b**) stream in the system. Circles indicate different species, grey lines indicate feeding interactions and the size of the circles is proportional to the population biomass of each species in the stream. Note the reduction in the number of consumer species in the food web for the warm stream, and the thinning out of feeding interactions compared with the cold stream. **c,d,** There was a reduction in mean trophic level (**c**; Linear regression: $F_{1,12}=16.10$, $P<0.001$, $r^2=0.54$; Equation of line: $y=1.536-0.0054x$) and directed connectance (**d**; Linear regression: $F_{1,12}=18.93$, $P<0.001$, $r^2=0.58$; Equation of line: $y=0.220-0.0023x$) as stream temperature increased (definitions of food-web properties are included in the Methods). Solid and dashed lines are the linear regression model fitting and 95% confidence intervals, respectively.

stronger links⁹. Similar patterns were obtained when the same streams were sampled again during April 2009 (Supplementary Fig. 4a,b).

To assess how these structural changes altered energy flux through the food web, we calculated the lengths and angles of all pairwise consumer–resource links in $\log_{10}[\text{body mass}]$ and $\log_{10}[\text{abundance}]$ space²⁵ (Fig. 2a,b). The average pathway of energy flux through the food web was shorter in warmer streams (Fig. 2c,d) due to a reduction in mean link length as temperature increased (Fig. 2e). This supports H3 and indicates stronger feeding pressure in the warmer streams^{13,14}, with the abundance of resources suppressed relative to the abundance of their consumers (Supplementary Fig. 5b). A link angle of -45° means that resource biomass equals consumer biomass²⁵ (Fig. 2a) and the mean link angle became progressively smaller (that is, less negative) than -45° at higher temperatures (Fig. 2f). This indicates that the biomass of consumers was on average greater than the biomass of their resources in the warmer streams (Supplementary Figs. 5c and 6), as predicted in H4. Inverted biomass pyramids are promoted by stronger top-down control, generalist feeding, larger predators and higher trophic transfer efficiency^{21,26}, all of which have previously been shown to increase with stream temperature in the Hengill system^{12–14}. However, they can only persist if resources are replenished

rapidly enough to meet the metabolic demands of consumers¹², that is, the standing stock of resources is low but production is high enough to maintain consumer biomass through time. Such top-heaviness is increasingly documented in nature when consumer pressure or anthropogenic disturbances are especially powerful (for example, in marine fisheries), but these systems are less stable than their pyramidal counterparts^{21,27}. We found similar patterns for mean link angle, but no effect on mean link length in the April 2009 sampling data (Supplementary Fig. 4c,d), suggesting that effects of temperature on the latter during August 2008 should be treated with caution.

Our model accurately predicted a high proportion of empirically observed feeding interactions for high quality food webs¹⁶— $75 \pm 3.9\%$ (mean $\pm$ s.d.) across all 14 streams (Supplementary Fig. 7). This shows that the ADBM can be a useful tool for predicting ecological networks¹⁶—at least for size-structured aquatic ecosystems such as our study streams¹²—even when interaction data are limited, as was the case for most studies to date²⁸. Our empirical measurements of food-web structure and energy flux were also strongly correlated with the ADBM predictions, although deviation of the slope from the 1:1 line suggests that the model did not produce an accurate quantitative prediction of connectance (Supplementary Fig. 8).

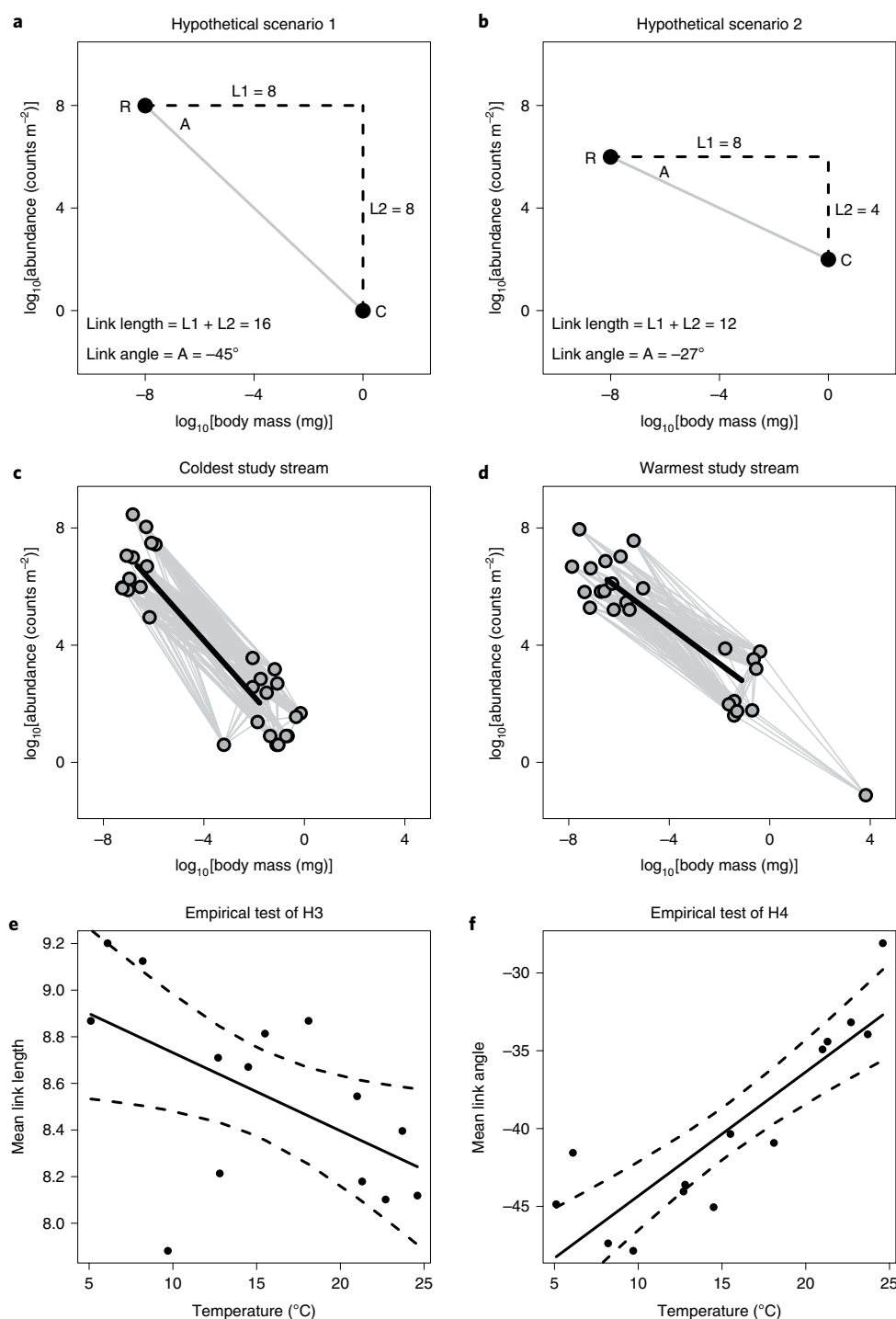


Fig. 2 | Temperature effects on energy flux. **a**, The length of a trophic link (grey line) is defined as the sum of the number of orders of magnitude of difference in body mass (dashed line labelled L1) and abundance (dashed line labelled L2) between a consumer (C) and a resource (R)²⁵. The angle (A) of a trophic link measures the rate of change in biomass from a consumer to a resource²⁵. Here, consumer biomass ($\text{mass} \times \text{abundance} = 10^0 \times 10^0 = 1 \text{ mg m}^{-2}$) equals resource biomass ($10^{-8} \times 10^8 = 1 \text{ mg m}^{-2}$), resulting in a link angle of -45° . **b**, A decrease in resource abundance and an increase in consumer abundance (relative to **a**) results in a shorter link length and a less-negative link angle. In this case, consumer biomass ($10^0 \times 10^2 = 100 \text{ mg m}^{-2}$) is greater than resource biomass ($10^{-8} \times 10^6 = 0.01 \text{ mg m}^{-2}$), resulting in a link angle of -27° . **c,d**, Trivariate food webs for the coldest (**c**) and warmest (**d**) stream in the system. Circles indicate species, grey lines indicate feeding interactions and the thick black lines represent the mean link length and mean link angle of the food web. **e,f**, There was a reduction in mean link length (**e**; Linear regression: $F_{1,12} = 5.04$, $P < 0.001$, $r^2 = 0.24$; Equation of line: $y = 9.067 - 0.0335x$) and a smaller mean link angle (**f**; Linear regression: $F_{1,12} = 37.28$, $P < 0.001$, $r^2 = 0.74$; Equation of line: $y = -52.30 + 0.797x$) as the temperature of the stream increased. Solid and dashed lines are the linear regression model fitting and 95% confidence intervals, respectively.

Our results indicate that the ADBM can also predict the impacts of temperature on natural food webs, using simple information that is routinely collected in ecological field studies. Further testing of this

model using other highly resolved food-web datasets from experiments that include manipulated warming in a controlled manner could validate this suggestion more broadly.

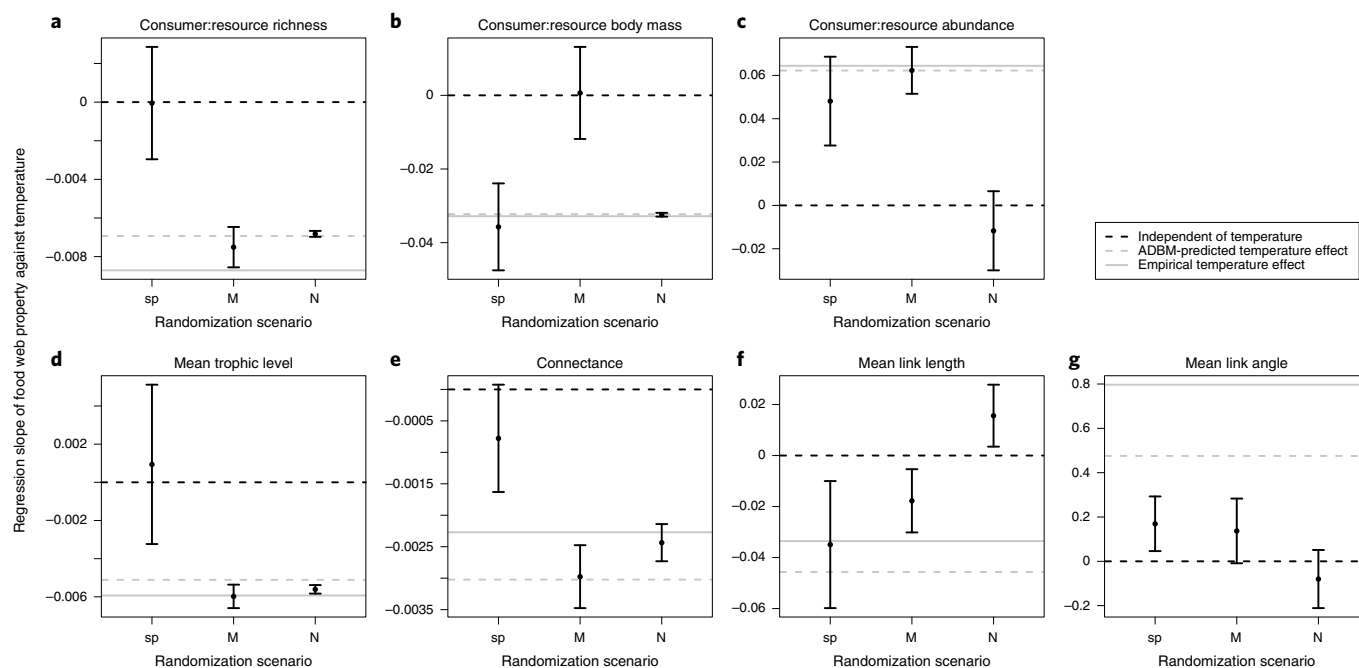


Fig. 3 | Effect of randomizations on temperature dependence of food-web properties. **a–c**, Tests of randomization success. **a**, The S scenario randomizes the species found in a stream and thus the ratio of consumer to resource species richness. **b**, The M scenario randomizes the average body mass of species in the stream and thus the ratio of consumer to resource body mass. **c**, The N scenario randomizes the population abundance of species in the stream and thus the ratio of consumer to resource abundance. **d–g**, Effects of randomizations on the properties of the food webs. The effect of temperature on mean trophic level (**d**) and connectance (**e**) is removed in the S scenario. The effect of temperature on mean link length (**f**) and mean link angle (**g**) is removed in the N scenario. In all other cases, even if the randomization scenario disrupts the empirical and ADBM-predicted patterns, it maintains the directionality of the temperature effect on the food-web property. For **a–g**, data are mean $\pm$ s.d. of linear regression slopes between food-web properties and stream temperature for 1,000 randomizations. The black dashed line represents a regression slope of zero between a food-web property and temperature (that is, the property is independent of temperature) and the solid and dashed grey lines represent the empirical and ADBM-predicted regression slope of each food-web property versus temperature, respectively.

As a final exploratory step, we investigated the underpinning mechanisms by using the ADBM to simulate food webs after changing one of the three major input variables—species identity, average body mass and population abundance. By randomly choosing species from the regional species pool (S scenario), we disrupted the trophic structure of any given stream and thus the relationship between stream temperature and the ratio of consumer to resource species richness (Fig. 3a). By randomly choosing a mean body mass (M scenario) or population abundance (N scenario) for each species from the same trophic groups in the regional dataset, we disrupted the relationship between stream temperature and the ratio of consumer to resource body mass or abundance, respectively (Fig. 3b,c). For each scenario, we then simulated 1,000 food webs for each of our 14 study streams after randomizing one input variable and fixing the values of the other two variables as close to the real stream as possible (see Methods).

Our S scenario removed the effect of temperature on mean trophic level and connectance (Fig. 3d,e) with negligible effects of the other two scenarios. This suggests that the relative biodiversity of consumers and resources is a key determinant of these food-web properties. We used the 14,000 food webs simulated under the S scenario to explore this effect—-independent of temperature—and found that both mean trophic level and connectance increase with the ratio of consumer to resource species richness (Fig. 4a,b). Thus, the disproportionate loss of consumer species, which is widely predicted in response to warming^{6–8}, should lead to reductions in these food-web properties.

Although all three randomization scenarios disrupted temperature effects on link lengths and angles, our N scenario had the greatest effect by far (Fig. 3f,g), suggesting that the ratio of consumer to

resource abundance is the principal determinant of energy flux. We used the food webs simulated under the N scenario to explore this effect— independent of temperature—and found that link lengths and angles decreased in size as consumers approach the abundance of their resources (Fig. 4c,d). Thus, stronger top-down control that alters the shape of trophic-abundance pyramids, which is often reported in warmer environments^{3,4}, will suppress energy flux through the food web.

Our study shows systematic impacts of temperature on wild food webs (see also ref. ²⁹). Most riverine ecosystems in Europe and North America fall within the studied temperature gradient³⁰ of 5–25°C; our results should therefore be indicative of changes in food-web structure due to future warming within this range. Our findings highlight the importance of monitoring species interactions for successful management of ecosystems³¹, given that trophic structure is so sensitive to environmental change. For example, mean trophic level is increasingly used in the management of fisheries to identify overfishing at the top of the food web³² and connectance is a useful indicator of resistance to invasion³³ and robustness against biodiversity loss^{23,24}. We identified changes in the relative biodiversity or abundance of consumers and resources at higher temperatures as key mechanisms that drive the observed effects. Such changes are also elicited by anthropogenic activities—such as overexploitation and habitat degradation^{32,34}—emphasizing how the structure and stability of ecological networks may be threatened by a host of stressors. The predictive power of our model shows how the impact of these stressors could be anticipated and ultimately mitigated more broadly. These findings now need to be tested in a range of food webs from marine, freshwater and terrestrial realms to gauge their potential universality.

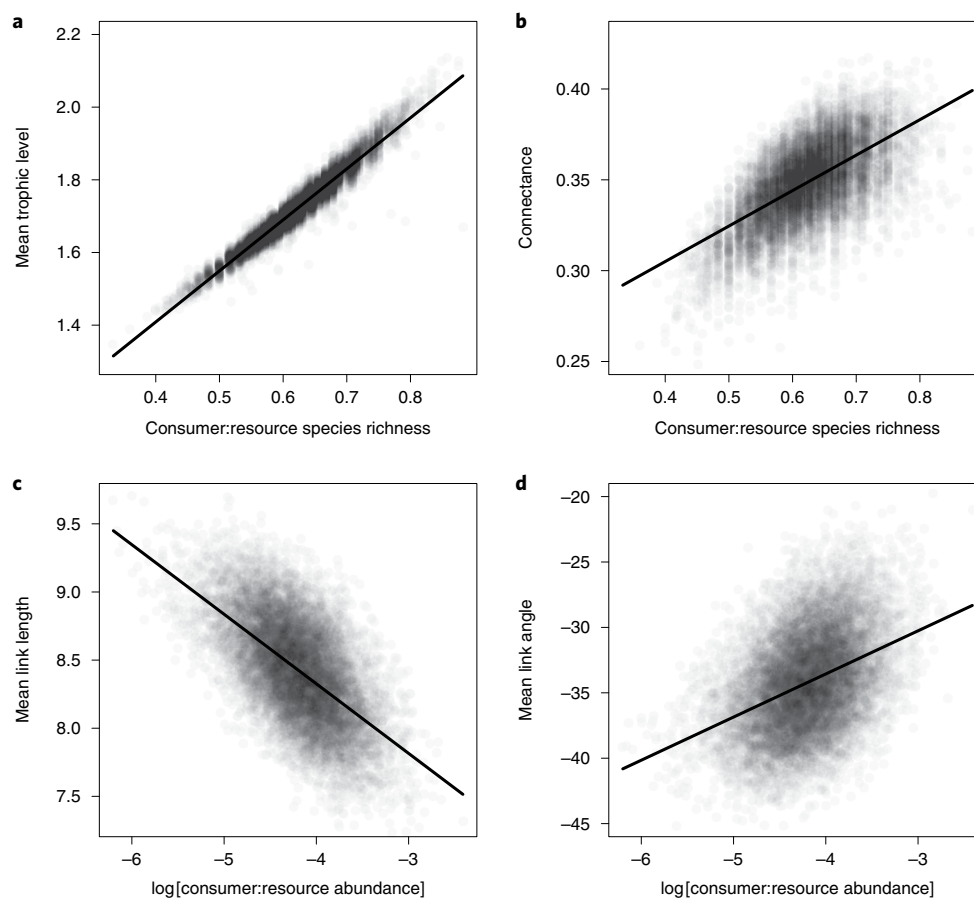


Fig. 4 | Key determinants of food-web properties. **a,b**, The effect of the ratio of consumer to resource species richness (independent of temperature) on mean trophic level (**a**; $y = 0.847 + 1.4044x$, $r^2 = 0.94$) and connectance (**b**; $y = 0.227 + 0.1952x$, $r^2 = 0.35$) in 1,000 food webs simulated for each of the 14 streams using the S scenario (compare with Fig. 3). **c,d**, The effect of the log-transformed ratio of consumer to resource abundance (independent of temperature) on mean link length (**c**; $y = 6.284 - 0.5105x$, $r^2 = 0.32$) and mean link angle (**d**; $y = -20.37 + 3.297x$, $r^2 = 0.13$) in 1,000 food webs simulated for each of the 14 streams using the N randomization scenario (compare with Fig. 3). For **a–d**, Parameter estimates are the mean intercept, slope and r^2 values from 1,000 linear regressions of the relationship across streams (that is, one regression for each randomization).

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of code and data availability and associated accession codes are available at <https://doi.org/10.1038/s41558-019-0513-x>.

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Author contributions

E.J.O., O.L.P. and G.W. were responsible for funding applications, research design and planning of the experiments. E.J.O., K.J.F., B.G., T.A.C.G., J.N.-C., J.S.O., D.E.P. and M.S.A.T. collected the data. E.J.O. and O.L.P. analysed the data. All authors wrote the paper.

Additional information

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Methods

Stream sampling. Streams were sampled during August 2008 and April 2009 to quantify the three major trophic groups in the system—benthic diatoms (three stone scrapes per stream), macroinvertebrates (five Surber samples per stream) and fish (three-run depletion electrofishing). Yield–effort curves were constructed to verify the efficiency of sampling¹³. We focus on the August 2008 data throughout because they represent the height of the growing season, whereas the April 2009 data are from a time of the year when the streams are in transition. Thus, we used only the latter to determine the consistency of the observed patterns over time. Diatoms and macroinvertebrates were identified at the species level under the microscope and counted to estimate population abundance, which was scaled to number of individuals per m² on the basis of sampling areas. Average body mass (in mg of dry weight) was estimated from linear measurements for at least ten individuals of every species and published length–weight relationships (Supplementary Tables 4 and 5). Note that diatoms could only be reliably identified to the genus level in gut contents, so we calculated the total abundance and abundance-weighted mean body mass of each diatom genus from the species-level data. Nevertheless, we refer to all taxa as species throughout this paper. Body-mass measurements of the only fish in the system (brown trout, *Salmo trutta*), were taken on a portable mass balance and converted to dry weight according to a wet-weight-to-dry-weight relationship¹². Precise details of the study system and stream sampling are given in the Supplementary Methods.

Overview of food webs. Direct observations of feeding links in nature are preferable to inferences based on indirect evidence, experiments or previous publications from other study sites²⁵. Nevertheless, food-web studies are plagued by undersampling of rare species and links when food webs are constructed entirely on the basis of direct observation, and by overestimation of links when food webs are entirely inferred from the literature^{28,35}. A yield–effort curve for links as a function of cumulative sampling effort should be reported for all direct observations³⁵, but this is still rarely the case in most food-web studies²⁸. Here we performed extensive gut-content analysis on organisms collected from our study system and used yield–effort curves to assess the completeness of our sampling effort. We supplemented the undersampled component with inferences from the literature to achieve the optimum balance between underestimation and overestimation of true food-web structure.

Gut-content analysis. We documented 49,324 feeding interactions from 1,128 individual consumers collected from the Hengill streams using gut-content analysis. We used three different approaches: stomach flushing of fish (5,856 interactions from 109 individuals), acid digestion of macroinvertebrates (25,105 interactions from 289 individuals) and dissection of gut contents (18,363 interactions from 730 individuals). Organisms flushed from fish stomachs were immediately stored in 70% ethanol and later identified under the microscope^{13,14}. Macroinvertebrates were immersed in 62% nitric acid at 65 °C for 18 h to remove all organic matter except for silicate diatom frustules, enabling accurate identification of diatoms³⁶—the major primary producers in the streams¹³. A 1 ml subsample of the resulting suspension of diatom frustules was pipetted onto a glass coverslip and allowed to dry before being fixed to glass slides by adding a drop of Naphrax on a 60 °C hotplate. We identified the first 100 diatoms (where possible) that we encountered in a continuous non-overlapping 100-µm-wide transect, following a fixed route across the slide; this was found to be sufficient to accurately characterize the species present on each slide³⁶. Dissection of gut contents allowed us to quantify predation on other macroinvertebrates and feeding interactions with basal resources other than diatoms, that is, coarse particulate organic matter (which is >1 mm), fine particulate organic matter (which is <1 mm), macrophytes, filamentous green algae, microscopic green algae, cyanobacteria and terrestrial subsidies. Invertebrates were dissected at ×20 magnification and the gut contents were mounted onto glass slides with Hoyer's medium. Gut contents were quantified in three randomly chosen fields of view at ×200 magnification on a compound microscope.

Yield–effort curves. We constructed yield–effort curves using the *fitspecaccum* function of the *vegan* package in R v.3.5.0. Our community dataset was a matrix in which rows represented unique consumer guts analysed, columns represented resource taxa and values indicated the number of times each resource taxon was observed in the gut of a consumer. We used *method = exact* and set *fit* equal to each of the following models: *arrhenius*, *gleason*, *gitay*, *lomolino*, *asympt*, *gompertz*, *michaelis-menten*, *logis* and *weibull*. We chose the best-fitting model according to the Akaike Information Criterion and used the *predict* function of the *stats* package in R to estimate the predicted number of resource taxa for each consumer. Here, we set the parameter *newdata* in the *predict* function as either 2 × the number of guts analysed for that consumer or 50, whichever was the larger value. We carried out this procedure for four different groupings of consumer diet: (1) every consumer species in each stream; (2) every consumer family in each stream; (3) every consumer species in the Hengill region; and (4) every consumer family in the Hengill region.

Food-web construction. To construct a food web for a given stream, we started with the species list obtained from the sampling of that stream during August 2008.

We then added links for each species from gut-content analysis of those species in that stream. If yield–effort curves suggested that <95% of the diet was described for any species (Supplementary Fig. 9), we added links for consumers in the same taxonomic family obtained from the gut-content analysis of those families in that stream. If yield–effort curves suggested that <95% of the diet was described for any family (Supplementary Fig. 10), we added links for each species obtained from the gut-content analysis of those species across all streams in the Hengill region. If yield–effort curves suggested that <95% of the diet was described for any species in the Hengill region (Supplementary Fig. 11), we added links for consumers in the same taxonomic family obtained from the gut-content analysis of those families across all streams in the Hengill region. If yield–effort curves suggested that <95% of the diet was described for any family in the Hengill region (Supplementary Fig. 12), we added links described for that species from previous studies (Supplementary Table 6). Only 12.6% of links were added from the literature, whereas 43.5% of links were directly observed from the target stream, and the remaining 43.9% of links were directly observed from the Hengill region. From our directly observed links, 74.3% of links were specific to each consumer species, whereas only 25.7% of links were inferred from the family level.

Food-web properties. Food webs were visualized and their properties were calculated using the *cheddar* package in R. The triangular food webs in Fig. 1a,b and the trivariate food webs in Fig. 2c,d were visualized using the *PlotWebByLevel* and *PlotMvN* functions, respectively. Mean trophic level was calculated using the *ShortWeightedTrophicLevel* function, which is the average of the shortest trophic level of a consumer and one plus the mean trophic level of all its trophic resources. This metric has been shown to closely approximate flow-based trophic level, for which each link is weighted according to its relative energetic contribution to the consumer's diet³⁷. Connectance was calculated using the *DirectedConnectance* function, which is the proportion of possible links in a food web that are realized³⁸. Mean link length and mean link angle were calculated from the *length* and *angle* columns using the *links* data frame returned by the *NvMTriTrophicStatistics* function. Link lengths describe the distance in mass–abundance space between every consumer and each of its resources in the food web, whereas link angles describe the biomass of every consumer relative to each of its resources (Fig. 2a). These metrics are increasingly used to quantify the flux and distribution of biomass throughout the food web^{25,39–42} and provide more-precise information than biomass pyramids, which only describe the total biomass at each discrete trophic level (Supplementary Fig. 6). The ratio of consumer to resource species richness was calculated as the number of consumer species divided by the number of resource species. The difference between the log₁₀-transformed abundance-weighted arithmetic mean body mass of consumers and of resources was taken as the log ratio of consumer to resource body mass. The difference between the log₁₀[mean abundance] of consumers and of resources was taken as the log ratio of consumer to resource abundance. The difference between the log₁₀[mean body mass × abundance] of consumers and of resources was taken as the log ratio of consumer to resource biomass. Temperature effects on food-web properties were analysed with linear regressions using the *lm* function of the *stats* package in R, with each food-web property taken in turn as the dependent variable and stream temperature as the explanatory variable.

ADBM. The ADBM is a model of food-web structure based on optimal foraging theory. It predicts the qualitative structure of real food webs, often to a high degree of accuracy¹⁶. By incorporating the temperature dependence of foraging traits, the model has also been shown to be a useful framework for predicting the effects of temperature on food-web connectance¹⁰. The ADBM predicts the diet *r* of each consumer *j* that maximizes the rate of energy intake:

$$\frac{\sum_{i=1}^r N_i a_{ij} E_i}{1 + \sum_{i=1}^r N_i a_{ij} h_{ij}} \quad (1)$$

where *N_i* is the density of resource species *i*, *a_{ij}* is the attack rate of consumer species *j* on species *i*, *E_i* is the net energy gained by consumption of an individual of species *i* and *h_{ij}* is the time taken for species *j* to handle an individual of species *i*.

The body mass and temperature dependence of *a_{ij}* can be described as:

$$a_0 M_i^{a_i} M_j^{a_j} e^{\frac{T-T_0}{kT_0}} \quad (2)$$

where *a₀* is a normalization constant for attack rate, *M_i* is resource body mass (in mg), *M_j* is consumer body mass (in mg), *a_i* and *a_j* are allometric exponents, *E_a* is the activation energy of attack rate (in eV), *T* is environmental temperature (in K), *T₀* sets the intercept of the temperature relationship at *T₀* rather than at zero Kelvin, and *k* is the Boltzmann constant (8.618 × 10^{−5} eV K^{−1}). The value of *E_i* is determined by the proportion of dry-to-wet mass in each organism^{43,44}, *ε_p* and may vary with temperature⁴⁵, but for simplicity we assumed here that it would be directly proportional to body mass in all streams^{10,16}, that is, *E_i* = *ε_p* *M_i*.

The body mass and temperature dependence of h_{ij} can be described as:

$$\frac{h_0}{h_b - \frac{M_i}{M_j}} e^{\frac{E_h(T - T_0)}{kTT_0}} \quad (3)$$

where h_0 is a normalization constant for handling time, h_b is a critical mass ratio and E_h is the activation energy of handling time (in eV). Note that $h_{ij} = \infty$ if $M_i/M_j \geq h_b$. We let $h_b = b_0 M_j^b$ because resource body mass has been shown to vary with consumer body mass according to a power law⁴⁶. Note that we let $b_0 = 1$ with dimensions that cancel those of M_j^b . We used a ratio-handling-time function in equation (3)—rather than a power-handling-time function—because the latter is generally shown to have weaker predictive power⁴⁶ and was found to be a poor predictor of empirical food-web structure in the Hengill streams. The values for all parameters used in this study are listed in Supplementary Table 7, see Supplementary Figs. 13–16 for an exploration of the sensitivity of key food-web properties to the chosen parameter values.

It is important to note that the estimates of food-web structure on the basis of the ADBM are independent of the empirical quantification of food-web structure using dietary analysis. The former relies exclusively on the body mass and abundance information for each species to determine food-web links, whereas the latter determines the links from direct observations from gut contents (>87% of cases) or inference from the literature. Thus, empirical measurements of mean trophic level and connectance are completely independent of the ADBM predictions of these metrics. Although mean link length and mean link angle incorporate body mass and abundance information, their values are determined by how consumers and their resources are distributed in mass–abundance space (that is, there is a major contribution of independent trophic link data to these metrics).

Randomization scenarios. We used the ADBM framework to simulate 1,000 food webs for each of our 14 study streams according to three different randomization scenarios. In the S scenario, we randomly selected n species from the regional species pool (where n is the number of species in a given stream) with the actual body mass and abundance for each species per stream, or the body mass and abundance from the stream that had the closest temperature in cases in which a species was not found in a stream. This scenario destroyed the ratio of consumer to resource species richness by changing the number of species belonging to each major trophic group (that is, diatoms, macroinvertebrates or fish) in each stream, but approximately maintained the ratios of consumer to resource body mass and abundance within each stream (Fig. 3a–c). In the M scenario, we maintained the species found in a stream and their population abundances in that stream but randomly chose body masses from the same major trophic groups in the regional species pool. This scenario destroyed the ratio of consumer to resource body mass, but approximately maintained the ratios of consumer to resource species richness and abundance within each stream (Fig. 3a–c). In the N scenario, we maintained the species found in a stream and their mean body masses in that stream, but randomly chose abundances from the same major trophic groups in the regional

species pool. This scenario destroyed the ratio of consumer to resource abundance, but approximately maintained the ratios of consumer to resource species richness and body mass within each stream (Fig. 3a–c).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from E.J.O. on reasonable request.

Code availability

The R code that supports the findings of this study is available from E.J.O. on reasonable request.

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Reporting Summary

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- ☐ ☒ Clearly defined error bars
State explicitly what error bars represent (e.g. SD, SE, CI)

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Software and code

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Data collection

All empirical data were collected in the field, as described in Methods. R packages used to construct yield-effort curves and food webs are described in the main text. All R code used to generate food webs using the allometric diet breadth model is available from the first author on reasonable request.

Data analysis

All statistical analyses were carried out in R 3.5.0.

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Ecological, evolutionary & environmental sciences study design

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Study description	Species composition, mean species body mass, total species abundance, and food web structure are described for 14 freshwater streams. Linear regression analyses were performed to quantify the relationship between food web properties and stream temperature. Food web properties were also predicted using a theoretical model, with empirically measured mean species body mass, total species abundance, and stream temperature as parameters. Linear regression analyses were again performed to quantify the relationship between predicted food web properties and stream temperature and between empirically measured and predicted food web properties. Input parameters to the model were randomised 1,000 times for each stream to help identify the mechanism driving the observed patterns.
Research sample	Fish, macroinvertebrates, and diatoms were sampled from 14 streams in August 2008, representing the major trophic groups in these freshwater ecosystems.
Sampling strategy	Yield-effort curves were constructed to verify the completeness of stream sampling (see Extended Data Table 1). Gut content analysis was performed on 1,128 individual consumers to obtain 49,324 directly observed feeding links. Yield-effort curves were again constructed to verify the completeness of sampling (see Supporting Information).
Data collection	Samples were collected and immediately preserved in the field for later identification in the laboratory. Researchers involved in the field collections and laboratory processing included Eoin J O'Gorman, Doris E Pichler, Katy J Faulkner, Bruno Gallo, Timothy AC Gordon, Joana Neto-Cerejeira, Murray S Thompson, and Jón S Ólafsson. All samples were identified using stereo microscopes (including a Nikon Eclipse 50i, an Olympus BH2, Nikon SMZ1500, and a CETI Vari Zoom 10) and the data were recorded in Excel spreadsheets, which were subsequently screened for errors, compiled, and managed by Eoin J O'Gorman.
Timing and spatial scale	Electrofishing, surber samples, and rock scrapes were collected from the 14 study streams between 1st and 7th August 2008. Additional macroinvertebrates for stomach content analysis were collected from 19th April to 12th May 2009, from 6th to 28th September 2012, from 1st-26th July 2013, and from 17th May to 7th July 2017. All study streams were located within 2 km of each other in the Hengill geothermal valley, Iceland.
Data exclusions	No data were excluded from any of the 14 streams that were analysed in this study. We excluded two other streams from the Hengill geothermal valley that have been examined in some previous publications: (1) IS15 was excluded because it has a temperature of 40-50 °C, representing an extreme environment dominated by cyanobacteria mats and only one thermophilic invertebrate species, <i>Scatella thermarum</i> ; (2) the benthos of IS2 is composed entirely of soft sediment, making it impossible to sample diatoms in a way that would be comparable to the other streams in the system (via stone scrapes).
Reproducibility	We collected additional samples from the 14 study streams in April 2009 and August 2012, which will allow us to verify the consistency in our patterns in different years and seasons. We have also repeated the methodology in other geothermal systems around the Arctic Circle (in Greenland, Alaska, Svalbard, and Kamchatka; see O'Gorman et al. 2014 in <i>Global Change Biology</i>). We are in the process of analysing these data for further papers.
Randomization	This study involved a survey of a natural warming experiment, consisting of 14 streams of different temperatures. As such, there was no allocation of samples/organisms/participants into groups and no need for randomisation.
Blinding	Samples were allocated alphanumeric codes, which helped to avoid any observer bias during sample processing, i.e. the person did not know if they were processing a sample from a cold or warm stream and thus could not let any expectations about what they might find there influence their data collection.
Did the study involve field work?	☒ Yes ☐ No

Field work, collection and transport

Field conditions	Sampling of the streams in August 2008 was carried out over a one week period with minimal rainfall and air temperatures of approximately 20-25 °C.
Location	Fieldwork was performed in the Hengill geothermal valley, Iceland (N 64°03'; W 21°18') at 350-420 metres above sea level.
Access and import/export	Access to the field site was obtained through collaboration with local researchers from the University of Iceland (Prof Gísli Már Gíslason) and the Marine and Freshwater Research Institute (Dr Jón S Ólafsson). Samples were processed in laboratories at the University of Iceland where possible and otherwise shipped from Iceland to London. No permits were needed to ship macroinvertebrate or diatom samples (preserved in 70% ethanol and Lugol's solution, respectively).

Disturbance

The study caused minimal disturbance to the field site. Researchers were careful to stick to established walking trails and to avoid excessive trampling of the vegetation or streams during sampling. The sampling protocols were non-invasive, involving electrofishing (3 passes of a 50 m reach), surber sampling (5 samples per stream), and rock scrapes (3 samples per stream). Additional hand collections of macroinvertebrates were carried out to boost sample sizes for gut content analysis.

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involvement in the study
☒	☐ Unique biological materials
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals

No live animals were brought to the laboratory.

Wild animals

The only fish in the study streams, brown trout, were sampled by electrofishing. They were immediately weighed on a portable balance and a fork length measurement was taken. All individuals greater than 8 cm fork length were stomach flushed by injecting water into the stomach through plastic tubing that was gently placed through the mouth and esophagus. All fish were returned to the stream immediately after sampling.

Field-collected samples

Samples were collected and immediately preserved in the field for later identification in the laboratory. Fish stomach contents and macroinvertebrate samples were stored in 70% ethanol. Diatom samples were stored in Lugol's solution. Samples were stored in laboratories at the University of Iceland, Queen Mary University of London, and Imperial College London. Sample containers were stored in cardboard boxes to prevent contact with direct sunlight and room temperature did not exceed 25 °C throughout the storage period.